Combat Robot Optimized Weapon Spinner (CROWS)

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ROVING ROBOTICS

Introduction

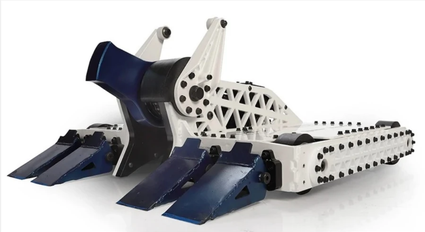

Fig. 1: Battlebots Champion "Bite Force" using a vertical spinner.

In combat robotics, spinning weapons store kinetic energy before each impact. Across hobby weight classes (1 lb antweight to 30 lb featherweight), weapon masses range 0.1–4 kg at 3,000–12,000 RPM, storing 50–10,000 J per hit.

Current practice relies on **empirical design rules**: wall-thickness proxies, manual iteration, and build-and-break testing. No principled framework simultaneously trades off geometry, structural safety, and engagement quality.

CROWS closes this gap with a modular optimization pipeline: physics-based objectives, 2D FEA-in-loop, and a GP surrogate for fast design screening.

System Architecture

Five independently swappable optimization slots:

Slot	Options	Config key
Outer profile	Fourier; B-spline; Bézier; Catmull-Rom	`profile_type`
Interior voids	Superellipse pockets; SIMP topology	`cutout_type`
Objective	Baseline (formula); enhanced (spiral+FEA)	`eval_mode`
Normalisation	None; robust rolling IQR	—
Eval gate	Off; staged (geom / cheap / FEA)	`staged_gate`

Design requirements:

- Maximise $\frac{1}{2}I\omega^2$ within mass & envelope budget
- Maximise kinematic bite depth per tooth engagement
- Safety factor $\mathrm{SF} \geq 1.5$ under centrifugal + contact loads
- CoM offset $\leq 3\%$ of $R_{\max}$ (dynamic balance)

Profile Parametrisation Methods

Five families map N=12 control radii to a closed boundary polygon. **Fourier radial** (baseline) and **periodic B-spline** (C^2) suffer global support and ill-conditioned Jacobians ($\kappa \sim 10^5$ and 10^{12} respectively), causing parameter directions invisible to gradient-free mutation. **Composite Bézier** and **centripetal Catmull-Rom** (both C^1) achieve local support, full rank, and κ<100; Catmull-Rom interpolates control points exactly, enabling archetype warm-starts.

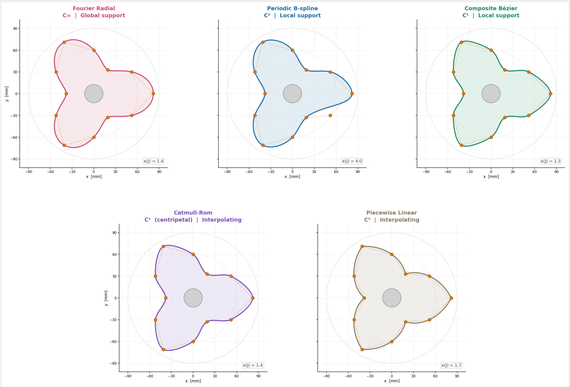

Fig. 2: All five families on a 3-lobe eggbeater (N=12, r_i=60+22 cos 3θ_i mm). Orange dots = control points; Catmull-Rom passes through each exactly. $\kappa(J)$ annotated per panel; B-spline and Fourier show elevated condition.

Functional Harmonic Parametrisation

Radial harmonic series (6 parameters):

$$r(\theta) = R_0 + A_1\cos\big(n(\theta-\phi)\big) + A_2\cos\big(2n(\theta-\phi)\big) + A_3\cos\big(3n(\theta-\phi)\big) \quad (1)$$

Analytic seeding from bite target:

$$n^* = \frac{(v/\omega)\cdot 2\pi}{b_{\text{target}}} \quad (2)$$

Two-stage functional optimizer:
Stage A — 6-dim DE in $\{n, R_0, A_1, A_2, A_3, \phi\}$, `popsize`=20. Rapidly explores the interpretable harmonic manifold.
Stage B — lifts to N=12 B-spline control radii warm-started from Stage A's lobed solution; proximity penalty $\lambda\|\mathbf{r}-\mathbf{r}_{\text{ref}}\|^2$ anneals (λ halved every stagnation plateau) allowing refinement beyond the harmonic manifold while preserving physical intent.

Optimization & Evaluation Criteria

Baseline weakness 1 — Formula bite: $b = v/(n_{\text{teeth}} f_{\text{rot}})$, n_{teeth} hardcoded per weapon style. Optimizer *cannot* reward sharp geometry.

Kinematic Archimedean spiral bite: opponent traces $r_{\text{enemy}}(\theta) = r_{\text{start}} - (v/\omega)\,\theta$. Count zero-crossings with weapon profile per revolution:

$$b_{\text{eff}} = \frac{(v/\omega)\cdot 2\pi}{n_{\text{contacts}}} \quad (3)$$

Smooth disk: n_c=1, b_{eff}≈1.7 mm. Twelve teeth: n_c≈12, b_{eff}≈0.14 mm.

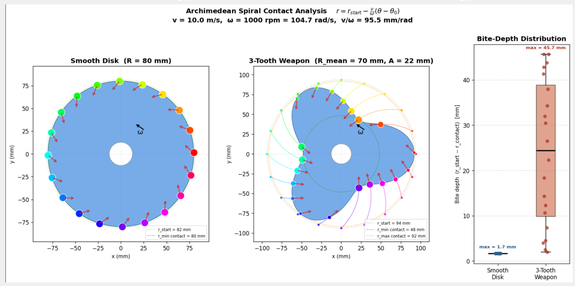

Fig. 3: Archimedean spiral overlays on smooth disk (left) and 3-tooth weapon (centre). Right: bite-depth polar roses. Max bite 1.7 mm (smooth) vs. 45.7 mm (toothed). Contact normal → FEA point-force loading.

2D CST FEA-in-loop (≈15 ms/call, Δx=10 mm):

$$\boldsymbol{f}_c = \rho\omega^2\boldsymbol{r}, \quad \mathrm{SF} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{VM,max}}}, \quad \mathrm{SF}<1.5 \;\Rightarrow\; E_{\text{score}} = 0 \quad (4)$$

Energy transfer objective:

$$E_{\text{transfer}} = \underbrace{\tfrac{1}{2}I\omega^2}_{\text{KE}}\cdot\underbrace{\cos^2\theta_c}_{\text{normal vel.}^2}\cdot\frac{b_{\text{eff}}}{b_{\max}}, \quad E_{\text{score}} = \frac{E_{\text{transfer}}}{E_{\text{ref}}} \quad (5)$$

FEA In-Loop — Optimization History (Heavyweight Disk):

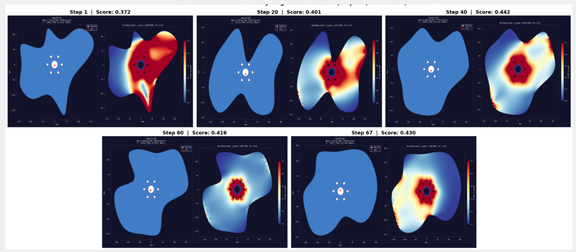

Fig. 4: Von-Mises stress maps at DE steps 1, 20, 40, 60, 67 (B-spline profile, 8,000 RPM, AR500 σ_y=1,400 MPa, colour normalised to yield). Profile evolves toward higher-MOI lobe geometry while avoiding stress concentrations at the bore and pocket boundaries.

Interior Weight-Relief Cutouts

Phase 2 superellipse pockets remove surplus mass while preserving structural integrity:

$$\left|\frac{x'}{a}\right|^n + \left|\frac{y'}{b}\right|^n \leq 1, \quad (x', y') = R_\phi\big(x - r\cos\phi,\; y - r\sin\phi\big) \quad (6)$$

Parameters (r, ϕ, a, b, n): radial distance, angular position, semi-axes, shape exponent.

Topology optimisation (SIMP) — alternative Phase 2:

$$E_{\text{eff}}(\rho) = E_{\min} + \rho^p(E_0 - E_{\min}), \quad p=3 \quad (7)$$

OC update minimises compliance minus weighted MOI under $\Sigma\rho_e A_e/A_{\text{total}} = V_f$. Density-filter radius $r_{\min}$=2.5 Δx suppresses checkerboard artefacts; hub and rim strips held solid.

FEA Solver Verification (Lamé analytical solution):

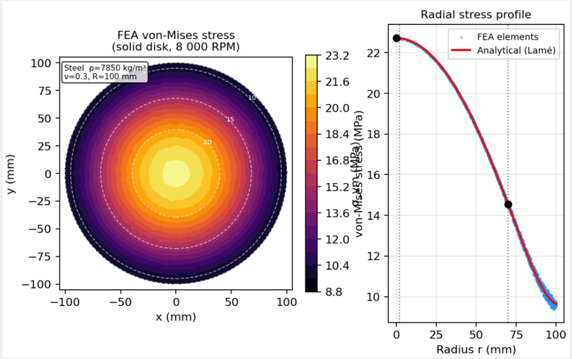

Fig. 5: Von-Mises field, per-element vs. Lamé profile, and error summary on a solid disk (8,000 RPM, AR500). Errors <2% at r=0.5R; <15% at r=0.7R. The 2D plane-stress CST mesh captures centrifugal stress accurately at combat-relevant speeds.

Stability Analysis — Profile Jacobian

Decompose $dO/d\boldsymbol{x} = J^\top\boldsymbol{q}$, $J = \partial\boldsymbol{p}/\partial\boldsymbol{x} \in \mathbb{R}^{360\times N}$. Low $\kappa(J) \Leftrightarrow$ uniform sensitivity: each control-point mutation moves the weapon boundary predictably.

Family	Cont.	Local?	$\kappa(J)$	rank	$\bar{c}$
PW-linear	C^0	Yes	95	12/12	1.000
Fourier	C^∞	No	1.1×10^5	10/12	1.000
B-spline	C^2	Part.	6.8×10^{12}	9/12	0.848
Bézier	C^1	Yes	76	12/12	1.000
Catmull-Rom	C^1	Yes	96	12/12	1.000

$\bar{c}$: gradient direction consistency across step sizes (1.0 = ideal). $\kappa\sim10^{13}$ for B-spline is **structural** — persistent for all $N \in \{4, \ldots, 20\}$.

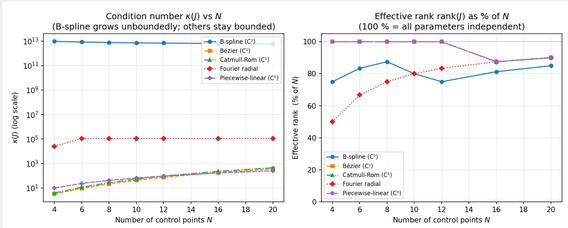

Fig. 6: $\kappa(J)$ and effective rank vs. N control points. B-spline ill-conditioning is structural — $\kappa\sim10^{13}$ at all N, with 15–25% of parameter directions invisible to mutation. Bézier and Catmull-Rom remain bounded and full-rank throughout.

Reduced-Order Surrogate Model

Motivation: Fine-mesh FEA ≈300 ms/evaluation. A GP surrogate targets ≈20× speed-up for dense design-space screening.

POD stress basis: $\tilde{\mathbf{S}} = \mathbf{U\Sigma V}^\top$; retain k modes (τ=95%: k=63; 99%: k=187). Modal coefficients trained via GP regression (one GP per mode). Safety-factor surrogate: single GP on scalar $\mathrm{SF} = \sigma_y/\sigma_{\text{VM,max}}$, enabling $P(\mathrm{SF}<1.5) = \Phi\left(\frac{1.5-\hat{\mu}}{\hat{\sigma}}\right)$ for Bayesian feasibility screening.

UCB acquisition: $a(\boldsymbol{\theta}) = \hat{E}_{\text{score}} + \beta\hat{\sigma}$

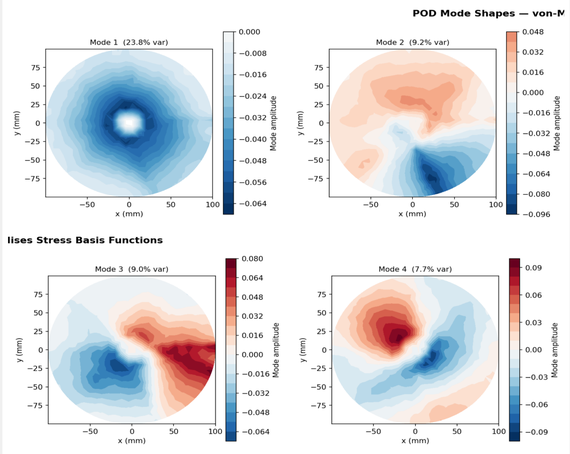

Fig. 7: First four POD stress basis functions (von-Mises, 2×2). Mode 1 captures 23.8% of stress-field variance; modes 1–4 collectively explain >50%, motivating the low-rank GP surrogate.

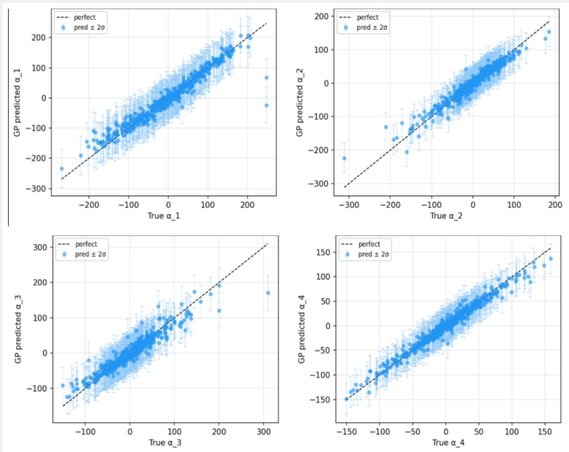

Fig. 8: GP predicted vs. true modal coefficient for modes 1–4 (trained on N=50 FEA designs): $R^2 \geq 0.906$. Calibration degrades beyond mode 30, motivating active-learning expansion toward constraint boundaries.

Conclusions

- **Catmull-Rom / Bézier:** $\kappa(J)$<100, full rank, $\bar{c}$=1.0 — preferred parametrisation for DE
- **Energy transfer objective** $E_{\text{transfer}} \propto I\omega^2\cos^2\theta_c \cdot b/b_{\max}$ unifies eval in one objectctive physical metric
- **Surrogate:** 3.9% LOO peak-stress error at N=50; UCB acquisition targets constraint boundary for efficient expansion
- **Future work:** Experimental validation of design; constraint-boundary seeding; improved ROM surrogate.